

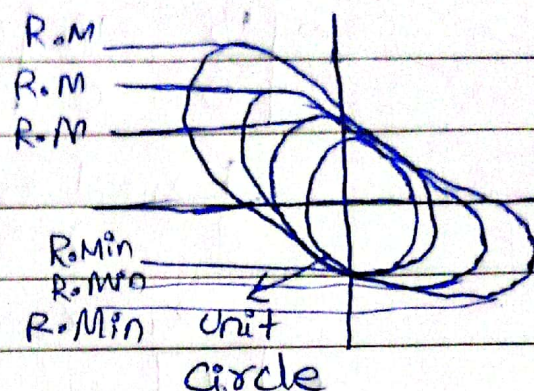
Problem 1:

$$5x^2 + 5y^2 + 4xy = 0$$

$$5x^2 + 5y^2 + 4xy = 1$$

$$5x^2 + 5y^2 + 4xy = 2$$

$$5x^2 + 5y^2 + 4xy = 3$$



making an ellipse. Plotted a unit circle.

Relative Maxima/minima are identified by observing the graph of curve. Highest & lowest points relative to each curve.

Problem 2:

$$\frac{3}{2}x^2 + \frac{3}{2}y^2 - xy = 4$$

$$\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 3/2 & -1/2 \\ -1/2 & 3/2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$x^T A x$$

$$A = \begin{bmatrix} 3/2 & -1/2 \\ -1/2 & 3/2 \end{bmatrix}$$

$$|A - I\lambda| = \begin{vmatrix} 3/2 - \lambda & -1/2 \\ -1/2 & 3/2 - \lambda \end{vmatrix}$$

$$(3/2 - \lambda)^2 - \frac{1}{4} = 0$$

$$\frac{9}{4} + \lambda^2 - 3\lambda - \frac{1}{4} = 0$$

$$\lambda = 2, \lambda = 1$$

$$\lambda^2 - 3\lambda + 2 = 0$$

$$\lambda = 1$$

$$\begin{bmatrix} 1/2 & -1/2 & | & 0 \\ -1/2 & 1/2 & | & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1/2 & -1/2 & | & 0 \\ 0 & 0 & | & 0 \end{bmatrix}$$

$$\frac{1}{2}x_1 = \frac{1}{2}x_2$$

$$x_1 = x_2$$

$$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix} x_2$$

$$\lambda = 2$$

$$\begin{bmatrix} -1/2 & -1/2 & | & 0 \\ -1/2 & -1/2 & | & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} -1/2 & -1/2 & | & 0 \\ 0 & 0 & | & 0 \end{bmatrix}$$

$$x_1 = -x_2$$

$$x = \begin{bmatrix} -1 \\ 1 \end{bmatrix} x_2$$

$$P = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$D = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

$$P^{-1} = \begin{bmatrix} -1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$$

$$x = Py$$

$$= x^T Ax$$

$$= (Py)^T A Py$$

$$= y^T P^T A Py$$

$$= y^T D y$$

$$= \begin{bmatrix} y_1 & y_2 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$= \begin{bmatrix} y_1 & y_2 \end{bmatrix} \begin{bmatrix} 2y_1 \\ y_2 \end{bmatrix}$$

$$= [2y_1^2 + y_2^2]$$

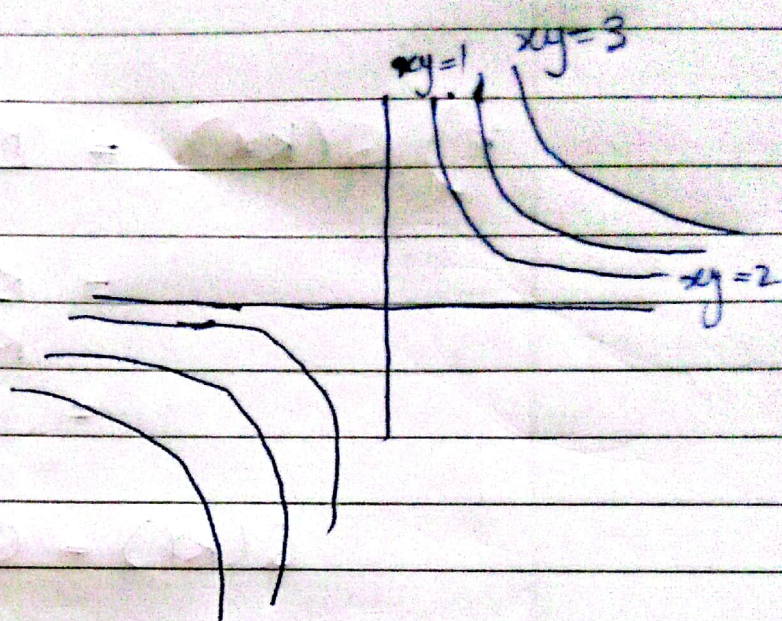
Problem 3:

$xy = 0$

$xy = 1$

$xy = 2$

$xy = 3$

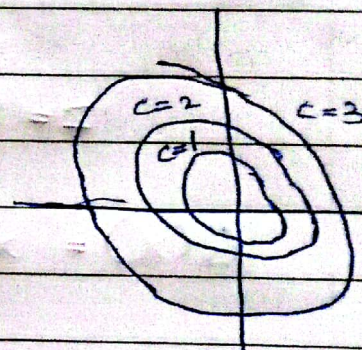
Problem 4:

$$\frac{3}{2}x^2 + xy + \frac{3}{2}y^2 = 0$$

$// = 1$

$// = 2$

$// = 3$

Problem 5:

i) $Q(x) = x^T A x$

$$Q^T(x) = (x^T A x)^T$$

$$= x^T A^T x$$

$$= x^T A x$$

$\because A$ is symmetric

$$Q(x) = Q^T(x)$$

$$\begin{aligned}
 \text{ii)} \quad Q(kx) &= \frac{(kx)^T A (kx)}{(kx)^T kx} \\
 &= \frac{k^2 x^T A x}{k^2 x^T x} \\
 &= \frac{x^T A x}{x^T x}
 \end{aligned}$$

$$Q(kx) = Q(x)$$

Problem 6:

$$z = 5x^2 + 5y^2 + 4xy$$

$$Q(x) = x^T \begin{bmatrix} 5 & 2 \\ 2 & 5 \end{bmatrix} x$$

$$5x^2 + 5y^2 + 4xy = 1 \quad \text{as constrained to } x^2 + y^2 = 1$$

